

WiSe 2026 Presentation Proposal

Presenter Information

Name: Abia Eunice Ipoikung

Affiliation: African Institute for Mathematical Sciences (AIMS), Cameroon

Presentation Details

Presentation Title:

Efficient Fine-Tuning of Large Language Models for Mathematical Problem Solving Using LoRA and Meta-Prompting

Abstract

Mathematical reasoning remains one of the most challenging tasks for large language models (LLMs), requiring multi-step inference and structured intermediate reasoning. Recent prompting strategies, such as meta-prompting, improve inference-time reasoning by guiding models through structured problem-solving procedures without requiring worked examples. Furthermore, parameter-efficient fine-tuning methods like Low-Rank Adaptation (LoRA) provide a computationally efficient way to adapt large-scale models while avoiding the high cost of full fine-tuning.

In this work, we investigate combining meta-prompting and LoRA fine-tuning for improving mathematical reasoning on LLaMA-3.3-70B-Instruct. We conduct a controlled study on the MATH and GSM8K benchmarks by evaluating four LoRA adapters trained on standard solution sets versus meta-prompted reasoning traces. Our framework allows us to compare standard adaptation with meta-prompted reasoning traces. Results demonstrate that the proposed approach significantly reduces computational overhead, offering a parameter-efficient and scalable direction for LLM adaptation.

Biography

Abia Eunice Ipoikung is a Master's scholar in Data Science at the African Institute for Mathematical Sciences (AIMS), Cameroon. Her research focuses on natural language processing and mathematical reasoning in large language models. She is currently working as an AI Engineer Intern at the telemedicine company WASPITO. Previously, she completed a Master 1 in Cybersecurity and has extensive experience as a software developer and IT instructor.